

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (AIML) (Credit System) Degree Examinations

May - June 2023

20AM603 - NATURAL LANGUAGE PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT*

1. a) What is NLP? Explain the origin of NLP and its challenges. 10 L*2
 b) List and explain different levels of processing involved in NLP. 6 L2
 c) Brief about any four applications of NLP. 4 L2

2. a) Explain Paninian Grammar Framework. Also, identify different Karaka's in the following sentence in Hindi Language.
 "Maan Bachchi ko aangan mein haath se roti khilathi hain" 10 L3
 b) Write a note on the following:
 (i) Hierarchical grammar
 (ii) Important features of Indian languages 10 L2

3. a) Describe the general n-gram model using bigram model. Find the probability of the test sentences for the given training set corpus:
 Training Set:
 <s> I am chintu </s>
 <s> I like college. </s>
 <s>. Do chintu like college. </s>
 <s> Chintu I am. </s>
 <s> Do I like chintu </s>
 <s>. Do I like college </s>
 <s> I Do like chintu. </s>
 Test Set:
 <s> I like college.
 Do I like chintu. </s> 10 L3
 b) Describe the following:
 (i) Data – sparseness problem in n-gram model.
 (ii) Smoothing techniques to handle the above problem (i). 10 L2

Unit – II

4. a) Explain the working of two-step morphological passer. Write the finite state transducers for mapping English nouns. 8 L2
 b) Describe hybrid Tagger. 4 L2
 c) Write minimum edit distance algorithm. Apply the algorithm to compute minimum edit distance between the words.
 PEACEFUL and PAECFLU. 8 L3

5. a) Write an algorithm for simple basic top-down passer. Illustrate step by step passing of the sentence 'open the door' using the same. 8 L2
 b) Differentiate Top – Down and Bottom – up parsing. 4 L3

P.T.O.

- c) Parse the sentence "Sona drinks coffee with milk" using Earley parsing algorithm where the grammar rules are defined as given below.

$S \rightarrow NP VP$ $VP \rightarrow VP PP$
 $NP \rightarrow NP PP$ $VP \rightarrow Verb NP$
 $NP \rightarrow Noun$ $Verb \rightarrow \text{drinks}$
 $PP \rightarrow Prep NP$ $Prep \rightarrow \text{with}$
 $Noun \rightarrow \text{Sana / milk / cofee}$

8 L3

6. a) Explain the various approaches in NLU.
 b) Compare NLP and NLU.
 c) Brief about the approaches of machine translations in NLU.

8 L2

4 L3

8 L2

Unit – III

7. a) Describe any two classical Models of IR system with suitable examples.
 b) Describe the design features of Information Retrieval Systems.
8. a) Brief about any two alternative models of IR systems.
 b) Write a note on the following.
 (i) Wordent
 (ii) Framenet
 (iii) Stemmers
 (iv) POS tagger.

10 L2

10 L2

10 L2

10 L2

BT* Bloom's Taxonomy, L* Level, CO* Course Outcome, PO* Program Outcome.
